

“WHERE
TECHNOLOGY
MEETS
INSIGHT”



IPSC IT FEST

08 - 10
DECEMBER
2026

Rules and Regulations

1. Invitation is open to all IPSC member schools. Registration closes on 25th September 2026, by which date schools must forward an event-wise list of participants.
2. Complete documentation for each event, including the detailed judging rubrics, will be circulated to all registered schools by the first week of October. This documentation will serve as the definitive reference for all participating schools.
3. Teams design, build and bring their own robots. Robots are not provided, and shall not be built from scratch on campus on the day of the event.
4. The competition is open to students of classes 9 to 12.
5. Each school may send a maximum of 8 students, accompanied by 1 faculty advisor.
6. Participants must bring their own laptops. The Hackathon Challenge is conducted on participants' own machines.
7. Robotics teams must bring their own tools, spares and consumables.
8. Batteries must comply with the limits in each event's robot specification. Damaged or swollen cells will not be permitted on campus.
9. Destructive mechanisms are prohibited across all robotics events. No robot may be capable of damaging an opponent, the arena or its components.
10. Internet access is provided where an event requires it and withheld where it does not, as stated in that event's rulebook. Mobile phones may not be used during any competition round.
11. Any unfair practice results in immediate disqualification. Judges assess all performances; where a referee or quizmaster officiates, their decision is final.
12. Teams are expected to arrive by 5:00 p.m. on 7 December 2026. Alternatively, teams may arrive on the morning of 8 December 2026. Teams may depart after the Closing Ceremony on 10 December 2026, which is expected to conclude by 2:00 p.m.
13. Schools are requested to make their own travel arrangements to and from The Doon School, Dehradun.
14. The Host School will provide accommodation and meals for all participating teams and their accompanying faculty advisor who opt to stay on the school campus. Registration charges, as well as boarding and lodging expenses, will be applicable as per the IPSC Charter.
15. All participants must carry school identity cards; those without proper identification will not be permitted to compete. Team escorts are responsible for the conduct of their teams at all times; any breach of discipline will lead to disqualification, in which event the rules and norms prescribed by the host school shall be followed.

TERRA INCOGNITA

1. The robot explores an arena it has never seen, mapping the space while tracking its own position within it. No prior map is permitted and the layout is not released beforehand.
2. Partway round, a movable block must be identified and pushed; pushing only, no lifting or grabbing.
3. The robot must then return as closely as it can to the point it started from. Recognising where it has already been, and correcting for accumulated error, is the heart of the event.
4. Operation is fully autonomous with all processing onboard; no remote control or wireless commands are permitted. LIDAR is not allowed, while ultrasonic, IR, time-of-flight, encoders and cameras are. The event rewards software skill over expensive hardware.

JUDGING:

- accuracy of the generated map against the official reference
- accuracy of the robot's final reported position
- completion of the in-arena object task.
- navigation carries higher weight than mapping alone.

PARTICIPANTS: 2



**[SLAM -
AUTONOMOUS
MAPPING &
PATHFINDING]**

[AGV - AUTONOMOUS GROUND VEHICLE NAVIGATION]



1. The robot scans a field of objects belonging to three classes: hazardous, fragile and normal. Each class is identified by colour, and sorts them into the matching coloured bins.

2. A symbol displayed before the run alters the sorting rules: a class may be redirected to a different bin, or the order of collection reversed. The symbol changes between rounds.

3. Hazardous objects are time-critical once lifted and must be deposited before the limit expires. Fragile objects must not be crushed or deformed in handling.

4. Objects are carried through a corridor network.

5. On completing the sort, the robot returns to the parking zone and parks itself. The entire run is autonomous, with no human intervention once it has begun.

JUDGING:

- correct classification and placement
- correct compliance with the symbol rule
- condition of fragile items and timing on hazardous ones
- return to station and total time.

PARTICIPANTS: 2

HALF-LIFE



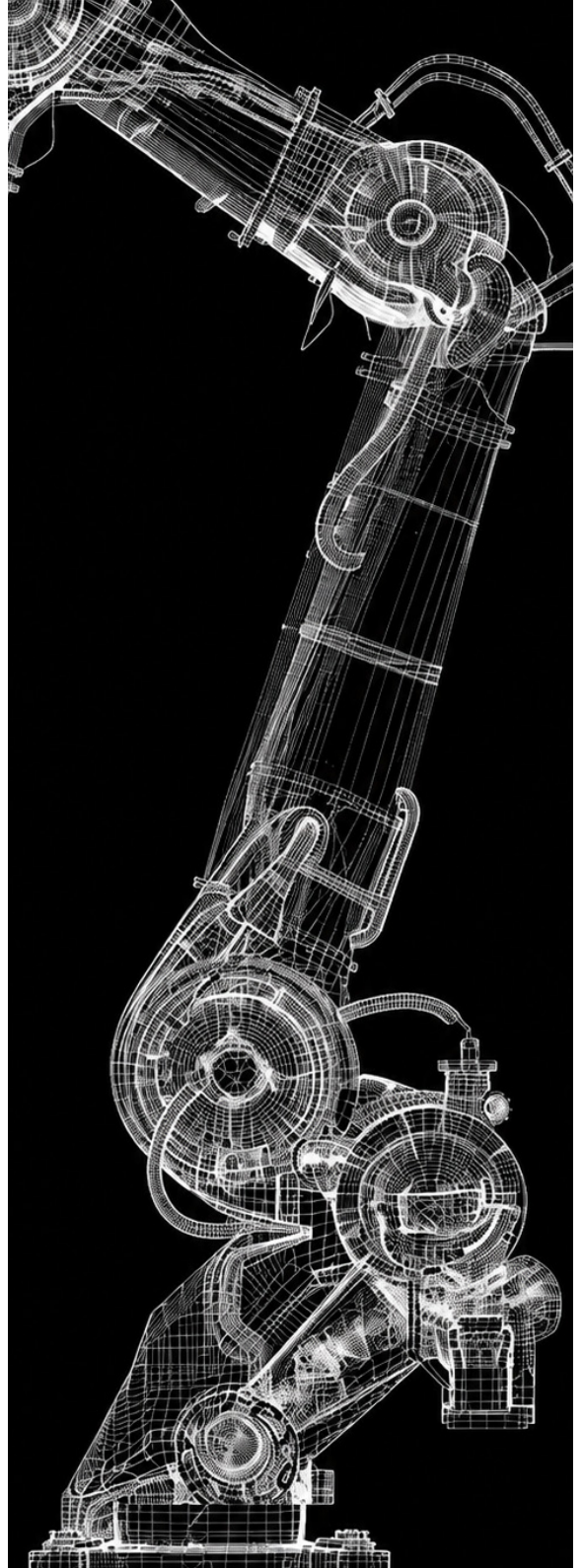
BUILD-DIFFERENT

1. A driver-controlled event fought across a quadrant-divided arena.
2. Stage one: retrieve the plants and deposit each into the pot of the matching colour.
3. Stage two: collect the pieces of the monument and assemble them in the prescribed configuration on the central platform. Full marks require the exact arrangement.
4. Teams reaching the semi-finals and finals face a third stage, disclosed only once qualifying has ended. It demands no change to the robot's hardware, but teams may reprogram. Build for adaptability, and bring a laptop.

JUDGING:

- correct colour-matching of every plant
- accuracy of the assembled monument
- driver precision with arena components
- total time

PARTICIPANTS: 2



**[TELE-OPERATED
ROBOTICS]**

[SUMO BOT]



1. Two robots face each other in a circular ring. The objective is to push the opponent out of it while remaining inside yourself.

2. Matches open with a fully autonomous phase, during which no driving is allowed; manual control is enabled for the remainder on the referee's signal.

3. A red band around the rim acts as a warning zone. A robot that enters it and fails to recover in time concedes a point to the opposition.

4. Destructive designs are prohibited: no blades, flame, liquids or projectiles, as are mechanisms generating artificial downforce, such as vacuum suction or sticky tyres. This is a pushing contest, not a combat one.

5. The event runs as a league, with the highest-scoring teams progressing to semi-finals and finals.

JUDGING:

- league results: push-outs.

PARTICIPANTS: 1

BRINKMANSHIP

HACKATHON

Participants will transform conceptual ideas into market-ready business solutions, combining technical expertise with entrepreneurial thinking.

The event will see the formation of corporate entities (“companies”), which will include multiple schools.

For example, five corporate entities may be established, each comprising five participating schools (with 15 members per company).

Please note that the total number of entities, schools per company, and total members per company may vary depending on total event registrations. The final numbers will be communicated with the registered school in September

Post-opening ceremony, companies must appoint a CEO, CTO, and CMO to oversee task delegation, steer strategic choices, and lead their companies.

The specific challenge/ problem statement will be released after the leadership selection.

From that point onward, companies will work until the final hour before the closing ceremony to:

- Design and develop a functional prototype (hardware, software, or a combination of both).
- Formulate a viable business and marketing strategy.
- Prepare and deliver a final pitch presentation.

Personal hardware (such as laptops) is required. While AI assistance is permitted for development and design, teams must demonstrate a profound understanding of their entire output.

JUDGING:

- 15-minute corporate pitch
- quality of functional prototype
- robustness of marketing strategy
- performance during the subsequent
- Q&A session.

**For hardware-based projects, participants will have access to campus fabrication resources, including 3D printing facilities, enabling them to rapidly prototype physical components where required.*

PARTICIPANTS: 3 (per school)





A team-based competition designed to test participants' knowledge of technology, computers, and the ever-evolving world of Information Technology.

Preliminary Round: Conducted in a designated room using individual digital devices. Teams answer a series of Multiple Choice Questions (MCQs) before the timer expires. Top-scoring teams advance to the Final Round.

Final Round: Finalists compete live on stage guided by the Quizmaster, featuring General IT Knowledge, Rapid Fire, Visual Identification, Audio/Video-Based Questions, Buzzer Round, and Surprise/Special Rounds.

JUDGING:

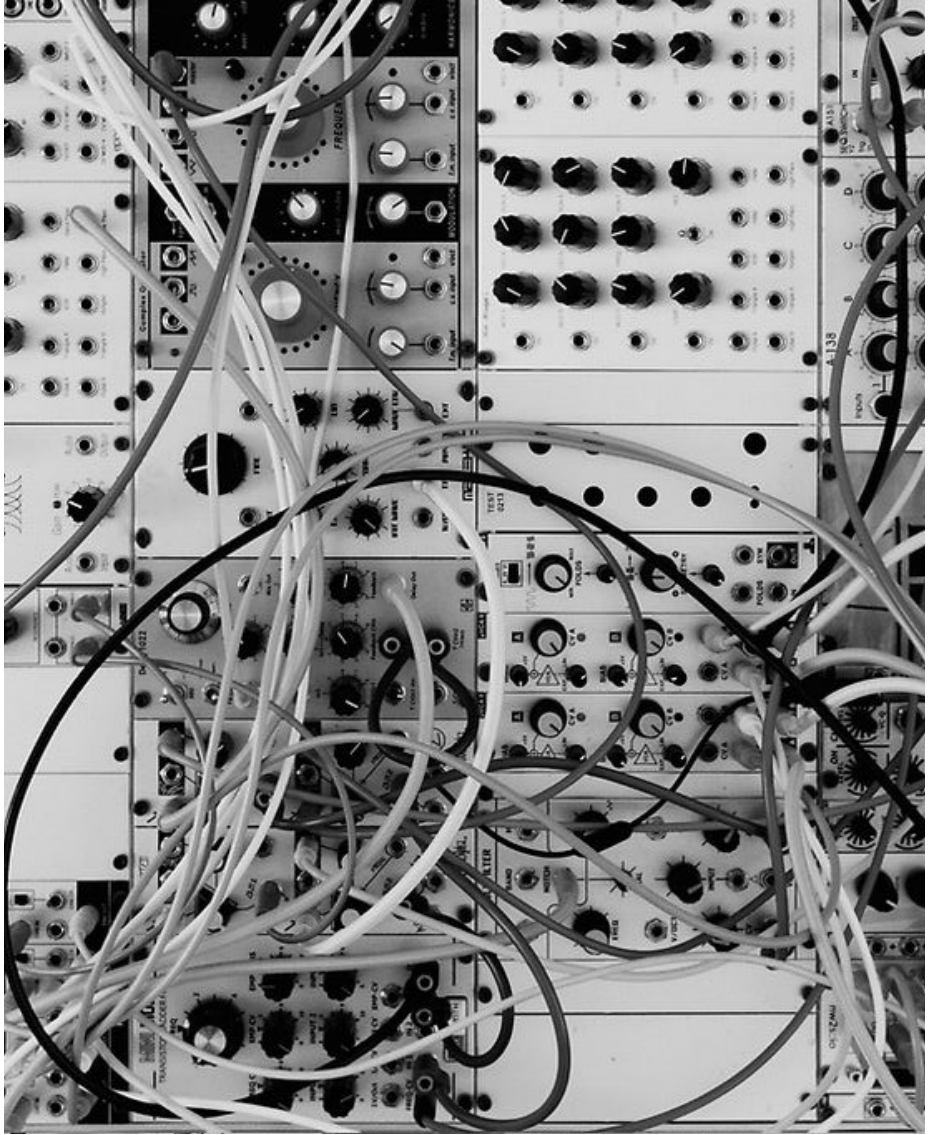
- accuracy of responses
- speed where applicable
- team coordination
- overall performance across all rounds.

Decisions made by the Quizmaster and judging panel shall be final.

PARTICIPANTS: 2

IT Q U I Z

ALGORITHM CHALLENGE



Participants demonstrate logical thinking, structured problem-solving, and clean coding practices.

Phase 1 (Design Before You Code): Every participant must prepare a solution design on paper using a flowchart, pseudocode, or step-by-step logic. Code implementation begins only after paper design is complete.

Phase 2 (Code Implementation): Implement the solution in Python, Java, C, or C++. Programs must be correct, efficient, well-structured, and capable of handling all valid test cases.

PARTICIPANTS: 1

JUDGING:

- submissions undergo three LeetCode-style passes
- weightage will be for both, algorithmic solution and actual implementation

Teams open a sealed envelope with a deliberately broken image dataset, imbalanced classes, color/background bias, blurry shots, missing samples, hidden artifacts (stickers, hands, background cues). No prior access allowed.

Phase 1 (Autopsy): Audit the data on paper. Log class counts, duplicate angles, lighting issues, background leaks, missing samples. Write your failure prediction: where will the model break and why? Training starts only after submission.

Phase 2 (Baseline): Train the model in Google Teachable Machine. Record overall accuracy, per-class scores, confidence. Must work across all classes.

Phase 3 (Hunt): Judges hand out 30 unseen test images. Test each, fill a confusion matrix, tag every error (color bias, background bias, angle, occlusion, too few samples), tally them, name your top two failure modes.

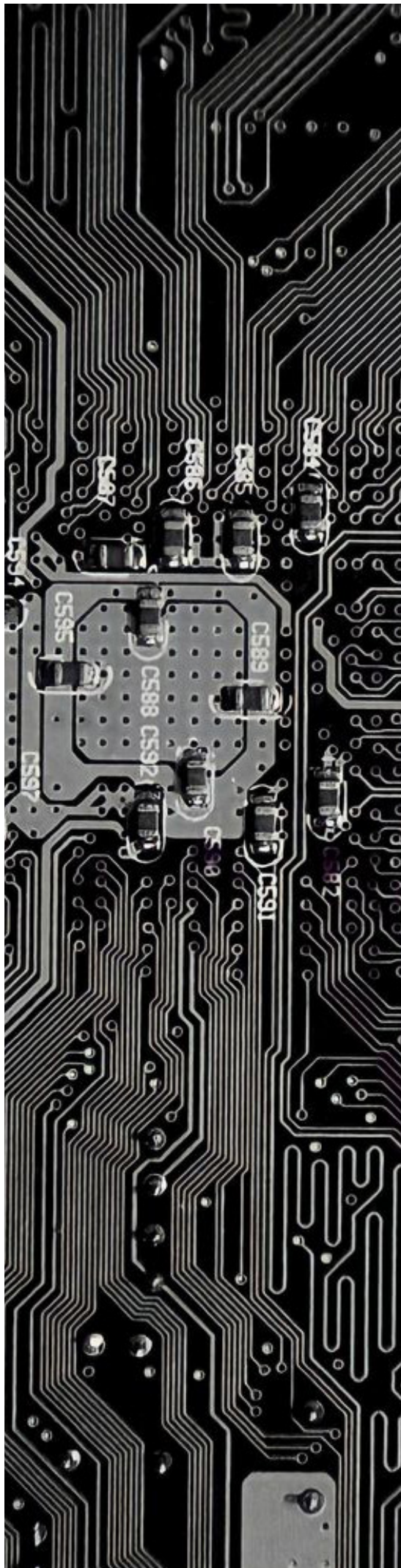
Phase 4 (Override): Shoot up to 10 targeted photos fixing those top two errors. Retrain. Re-test the same 30 images. Log hypothesis, fix, before/after numbers, trade-offs.

Phase 5 (Presentation): Walk us through what you found, what failed, what you fixed, what improved, what you learned. One judge question.

JUDGING:

- how well you audited, predicted, and planned
- how effectively the team identified and justified trade-offs
- accuracy on the hidden test set after your fix
- three review passes per team

PARTICIPANTS: 2



THE BLACK BOX

**TECHNOLOGY LIKE
ART IS A SOARING
EXERCISE OF
THE HUMAN
IMAGINATION.**

- DANIEL BELL



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